Mixed-Length Shapelets to KMeans

Picoclimate Example from Raw Data

Explainable Clustering for Spatio-Temporal Data

An Pham | LIRMM–ADVANSE | Supervisors: Luc & Reda

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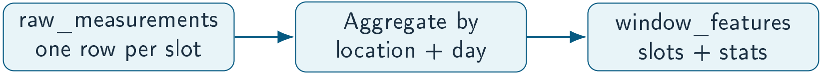
Raw Measurements to Window Features



Slot rows are aggregated into one daily window per location.

Example Row: raw_measurements.csv (Truncated)

timestamp	location	slot	air_temp	humidity	wind	pressure
2025-09-01 06:00	loc_000	morning	50.0	5.0	17.226	969.0

- Each row is one location + one time slot (morning/noon/afternoon/night).
- These rows become a 4-slot daily window in `window_features.csv`.

Example Row: window_features.csv (Truncated)

location	date	present	air_temp_s1	wind_s1	air_temp_s2	wind_s2	true_regime
loc_049	2025-09-24	0.9348	50.0	12.516	50.0	12.045	traffic_peak

- Slot columns (`__slot_1..__slot_4`) are the shapelet input.
- Mean/std/trend columns exist but are not used in the shapelet pipeline.

Shapelet Input Vector (Picoclimate)

- Build one 1D vector per window using only slot columns.
- Example (truncated):

$x =$ [slot1: temp, humidity, wind, ..., slot2: temp, humidity, wind, ..., slot4: ...]

- This is **not** a single-variable series.
- It is a flattened multivariate window.

Ordering Matters: Variable vs Slot

Bad (variable-grouped)

`temp_s1, temp_s2, temp_s3, temp_s4, humidity_s1, humidity_s2, ...`

Good (slot-grouped)

`temp_s1, humidity_s1, wind_s1, ..., temp_s2, humidity_s2, wind_s2, ...`

Slot-wise ordering keeps local shapelets temporally natural.

Mixed-Length Shapelets to Distance Features

- Short lengths only (e.g., 2–6) for 4-slot windows.
- Semi-exhaustive candidates + pruning (low variance, duplicates, correlation).
- Each shapelet yields one feature: minimum distance to the window.

Distance mapping

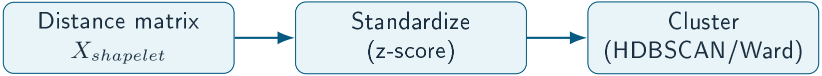
$$\phi_j(x) = \min_t \|x[t : t + L_j] - s_j\|_2$$

$$X_{shapelet} \in \mathbb{R}^{N \times m}$$

Candidate Strategy (Semi-Exhaustive + Pruning)

- Generate all candidates for short lengths on a subset of rows.
- Drop low-variance and duplicate shapelets.
- Keep top candidates by distance-dispersion score.
- Prune highly correlated candidates and keep a compact dictionary.

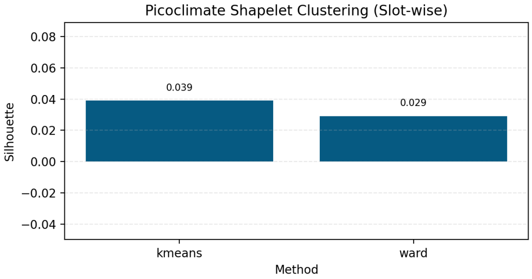
Clustering Input and Options



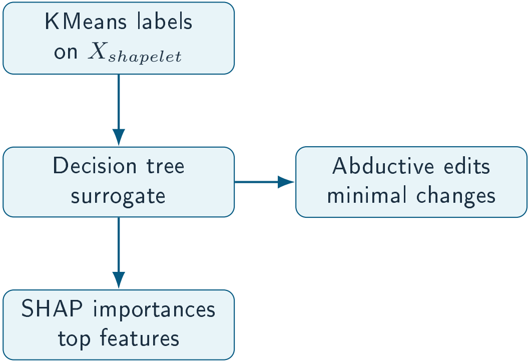
KMeans remains a baseline; HDBSCAN and Ward are primary options.

Picoclimate Summary (May 27, 2026)

- Config: max_rows=120, eval_rows=40, max_pool=1000, target_shapelets=200
- 1,500 windows; 87 slot columns; 188 shapelets
- KMeans best k=3 (silhouette 0.0391)
- Ward silhouette 0.0292 at k=3
- HDBSCAN unavailable in current env (install pending)



Explanation Stage (Transparent)



- Surrogate learns cluster rules in shapelet space.
- SHAP ranks which shapelet distances drive the label.
- Mean/std/trend features are used only for interpretation.
- Abductive edits show minimal changes to switch cluster.

Interpretation for T=4 Slots

- Shapelets act as local multivariate motifs, not long temporal patterns.
- Slot-wise ordering keeps motifs aligned to neighboring time slots.
- Statistics (mean/std/trend) provide global context after clustering.
- Single-variable shapelets are viable only with long sequences.